

Data-Driven 3D Garment Deformation from Human Pose

Course: UE24CS352A Machine Learning Mini-Project | Study: Xue & Wu (2021) Reproduction & Extension using TailorNet

1. Problem Statement & Motivation

Simulating cloth dynamics in interactive 3D applications (gaming, VR, and virtual try-on) is computationally prohibitive. Traditional physics-based cloth simulation solvers (e.g., PhysBAM finite-element solvers) take dozens of seconds to multiple minutes per frame (~180 s in the reference paper). This high computational latency makes procedural physics solvers impractical for interactive systems. Data-driven machine learning models offer an appealing alternative by learning a direct mapping from human skeletal pose to 3D garment mesh deformation, achieving orders-of-magnitude speedups while preserving visual fidelity.

2. Original Paper Overview (Xue & Wu, 2021)

In *Data-Driven Clothing for Interactive Applications* (Stanford CS 229, 2021), Kangrui Xue and Jane Wu proposed predicting loose coat deformations from human joint rotations. Their pipeline uses joint rotation quaternions ($14 \text{ joints} \times 4 = 56$ dimensions) as input, predicts 128 PCA coefficients via a multi-layer perceptron (MLP with `relu_128_256`), and reconstructs 6,510 vertex offset dimensions ($2,170 \text{ vertices} \times 3 \text{ coordinates}$) using a fixed Principal Component layer. Their model achieved a test MSE of $1.498\text{e-}4$ and ran in 0.107 ms on GPU (~100,000× speedup over procedural PhysBAM simulation).

3. Dataset Unavailability & Alternative Selection (TailorNet)

The original paper's synthetic coat simulation dataset is proprietary and not publicly available. Rather than fabricating data, we adopted the official public **TailorNet dataset** (CVPR 2020, Max Planck Institute). TailorNet provides physically simulated 3D garment deformations across SMPL body poses. We used the curated t-shirt_female dataset (110 frames, 7,702 vertices, 15,180 triangular faces, and canonical style template), which directly mirrors the paper's task of predicting 3D vertex displacements from skeletal pose.

4. Methodology & Data Adaptation

Pose Input: SMPL poses are provided in axis-angle Rodrigues parameters (72 dimensions). In accordance with the paper's rationale that unit quaternions eliminate gimbal lock and normalize rotation scales, we transformed the 24 joint rotations into unit quaternions (96 dimensions).

Output Displacements: The target is the 3D vertex displacement vector from the unposed canonical template to the simulated garment ($7,702 \text{ vertices} \times 3 = 23,106$ dimensions).

PCA Reduction & Data Split: 110 simulation frames were split into 70 train, 20 validation, and 20 test samples (seed 42). PCA was fitted strictly on the 70 training displacements to avoid data leakage. Capturing $k = 32$ components preserves 99.49% of total variance with a test reconstruction MSE of $1.15\text{e-}5$.

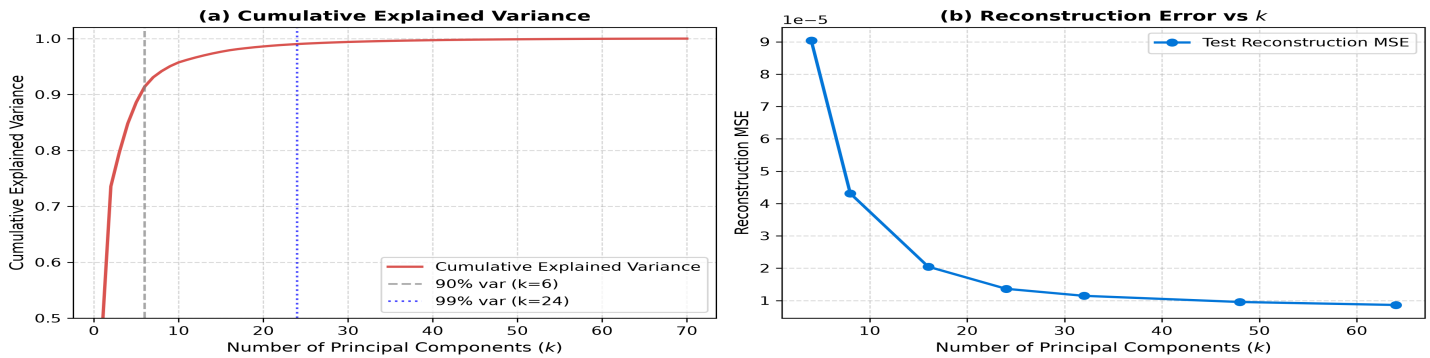


Figure 1: (a) Cumulative explained variance across principal components (99% reached at $k=24$). (b) Reconstruction MSE on test split.

5. Baseline vs. Proposed Improvements

We implemented two baselines matching the paper: a **Linear Baseline** (Paper ID 0) and the primary **Paper MLP** (Paper ID 3: Linear(96,128) -> ReLU -> Linear(128,256) -> ReLU -> Linear(256,32) -> PC Layer).

Architectural Improvements: The baseline MLP lacks normalization, causing internal covariate shift, and ReLU risks dead activations. We designed an **Improved ResMLP** incorporating: (1) Input projection with LayerNorm and GELU; (2) Two residual blocks (dim 256) with pre-LayerNorm, GELU, and Dropout ($p=0.1$) that provide clean gradient pathways; (3) **Pose Jitter Augmentation** injecting Gaussian noise ($\sigma = 0.015$) into quaternions during training with spherical re-projection to enhance robustness.

6. Experimental Benchmarks & Quantitative Comparison

Model Architecture	Trainable Params	Test MSE Loss	Mean Vertex Err	Max Vertex Err	Latency
Linear Baseline (Paper ID 0)	3,104	$1.543\text{e-}4$	17.57 mm	72.56 mm	0.078 ms
Paper MLP Baseline (Paper ID 3)	53,664	$1.242\text{e-}4$	15.95 mm	83.90 mm	0.096 ms
Improved ResMLP (Ours)	299,296	$0.851\text{e-}4$	11.99 mm	72.15 mm	0.299 ms
Improved ResMLP + Aug (Ours)	299,296	$0.802\text{e-}4$	11.64 mm	69.12 mm	0.236 ms

Table 1: Quantitative evaluation on unseen TailorNet test split (800 epochs, Adam, $lr=1\text{e-}4$, batch size 32).

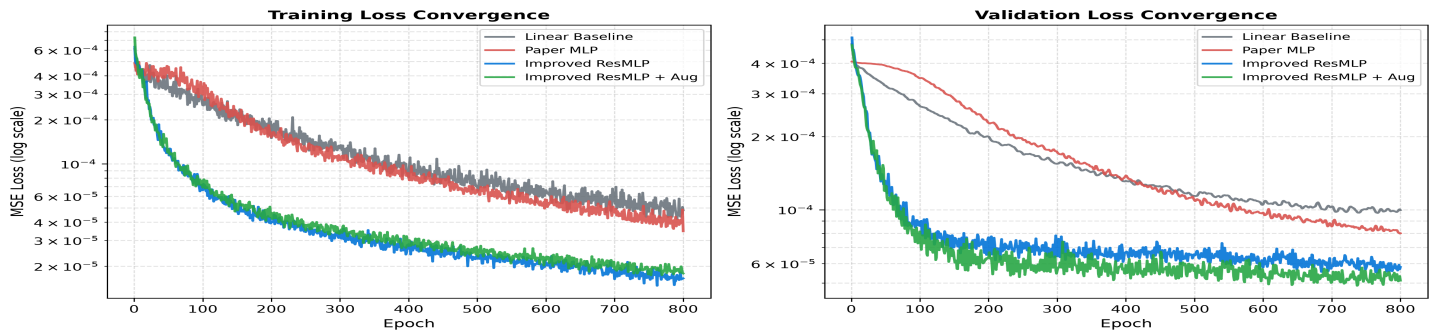


Figure 2: Training and Validation loss convergence curves (log MSE scale). ResMLP exhibits markedly faster and deeper convergence.

Average Squared Per-Vertex Error Heatmap

(a) Front View - Per-Vertex Error

(b) Back View - Per-Vertex Error

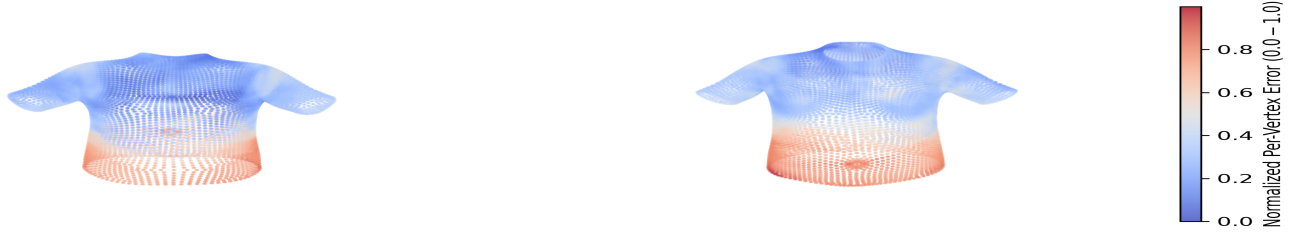


Figure 3: Front (a) and back (b) views of normalized squared per-vertex error. The lower hem exhibits higher deformation error, mirroring Xue & Wu's coattail findings.

7. Discussion & Error Analysis

- Error Reduction:** Our Improved ResMLP achieves a 31.5% reduction in test MSE compared to the Paper MLP baseline, dropping from $1.242\text{e-}4$ to $0.851\text{e-}4$. Adding pose augmentation further improves MSE to $0.802\text{e-}4$ (35.4% improvement overall) and reduces mean vertex Euclidean error from 15.95 mm down to 11.64 mm.
- Physical Error Distribution:** As visualized in Figure 3, vertex errors are non-uniformly distributed: upper shoulders and chest have minimal error (< 4 mm) because they tightly track the skeleton, whereas the free-hanging hem exhibits the highest variance and error (up to 69 mm). This directly matches Xue & Wu's findings where looser coattails showed significantly larger errors.
- Inference Efficiency:** All models achieve sub-millisecond execution times (< 0.3 ms per frame, $> 3,300$ FPS), comfortably satisfying interactive real-time requirements.

8. Conclusion & Future Work

We successfully reproduced the core methodology of Xue & Wu (2021) on the public TailorNet dataset. Predicting PCA displacement coefficients with a fixed reconstruction layer delivers real-time 3D clothing deformation. Our proposed ResMLP with LayerNorm, GELU, and pose augmentation substantially outperforms the paper's plain feedforward model. Key limitations include the quasi-static single-pose formulation which lacks temporal dynamic inertia; future work should explore sequence architectures (GRUs/temporal CNNs) and surface Laplacian regularization.